

SMARTFLOW

Multi-Object Tracking & Counting from Video

Technical Report and Two-Video Test Results

SMARTFLOW is a computer-vision system designed to detect, track, and count moving objects across video without double-counting. The implementation uses YOLO11 for object detection, ByteTrack for multi-object tracking, OpenCV for video processing, and a virtual line-crossing method for directional counting.

1. Project Objective

The system detects objects in each video frame and assigns a persistent tracking ID to each object. An object contributes to the count only when its tracked center point crosses the virtual counting line. Once an ID has been counted, it is stored in a counted-ID set so the same tracked object is not counted again. The dashboard also presents objects currently visible, cumulative crossings, direction, density, flow rate, and detected object classes.

2. Technology and Method

Component	Implementation	Purpose
Detector	YOLO11n	Detect objects and classify them
Tracker	ByteTrack	Maintain object IDs across frames
Video processing	OpenCV	Read, annotate and write video
Counting	Virtual line crossing	Count unique objects crossing the line
Dashboard	Streamlit	Display video, counts and analytics
Data output	CSV	Store frame-level tracking/count data

3. Counting Method

A vertical counting line is positioned at 60% of the frame width. For a video with width W , the line coordinate is calculated as $0.60 \times W$. For example1.mp4, the video width is 2160 pixels and the counting line is at $X = 1296$ pixels. The direction is determined from the previous and current X coordinates of the tracked object's center: left-to-right is recorded as a RIGHT crossing, while right-to-left is recorded as a LEFT crossing.

Objects that are detected and moving but never cross the line are not included in the crossing count. This prevents the same visible object from being counted on every frame. The counted-ID set provides the double-count prevention mechanism.

4. Video Test Results

Metric	traffic.mp4	example1.mp4
Video resolution	1920 × 1080	3840 × 2160
Frame rate	25 FPS	30 FPS
Frames	525	323
Counting line	60% width	60% width (X=1296)
System LEFT crossings	4	6
System RIGHT crossings	5	0
System total counted	9	6
Detected counted classes	Car, Truck, Person	Car, Person, Bus, Motorcycle
Average flow	25.71 objects/min	33.44 objects/min

5. Validation

example1.mp4: A manual review of the uploaded annotated video gave 6 LEFT crossings and 0 RIGHT crossings. SMARTFLOW also reported 6 LEFT and 0 RIGHT crossings. Therefore the manual total was 6, the system total was 6, the difference was 0, and the observed counting accuracy was 100%.

traffic.mp4: The latest SMARTFLOW processing run reported 4 LEFT crossings, 5 RIGHT crossings, and 9 total counted objects. A final manual validation for this run has not been established in this report, so an accuracy percentage is intentionally not claimed for traffic.mp4.

6. Dashboard and Outputs

The Streamlit dashboard supports switching between videos stored in the input folder. For the two-video test, the input folder contains traffic.mp4 and example1.mp4. The processing engine creates a video-specific annotated output and a video-specific CSV report, allowing results from multiple videos to be kept separately.

Example outputs include smartflow_traffic_tracked.mp4 and smartflow_example1_tracked.mp4, together with traffic_data.csv and example1_traffic_data.csv. The browser-friendly H.264 versions are used for dashboard playback.

7. Known Limitations and Failure Cases

The detector and tracker can occasionally miss objects, lose an ID during heavy occlusion, or assign a new ID after a long disappearance. These cases can affect counting accuracy. Small or distant objects may also be harder to detect reliably. The counting rule is designed for objects crossing the selected line; objects that remain on one side of the line are intentionally not counted.

YOLO11 uses its trained object classes. Some real-world road categories may not have a dedicated class in the model and can therefore be assigned to a visually similar class. This is a model-class limitation rather than a counting-line issue.

8. Conclusion

SMARTFLOW successfully demonstrates a complete multi-object detection, tracking, and line-crossing counting pipeline. Testing on traffic.mp4 produced 9 counted crossings, while example1.mp4 demonstrated multi-class tracking of cars, people, a bus, and a motorcycle with a manually verified 100% count match for that test. The system also provides CSV logging and a Streamlit dashboard for visualization.

9. Assignment Deliverable Status

Deliverable	Status
Working detect-and-track pipeline	Completed and tested on two videos
Manual vs system count report	Completed for example1.mp4; traffic.mp4 requires final manual validation
2–3 page technical write-up	This report
Dashboard with tracked video and live analytics	Completed
GitHub repository and README	Final update/push required